1.5 Continuous Functions

A function $f : R \to R$ defined on D is said to be continuous at $x \in D$ if, for any $\epsilon > 0$ there exists $d > 0$ such that when $x' \in D$ and $|x' - x| < d$, $|f(x') - f(x)| < \epsilon$.

A function f defined on D is said to be **continuous** if it is continuous at every $x \in D$.

Proposition 1.5.1: A function f is continuous at x, if and only if, for any convergent sequence $x_n \to x$, its image sequence converges to $f(x)$, i.e. $f(x_n) \to f(x)$. *Proof:* Suppose f is continuous at x, and $x_n \to x$. For any ϵ, there exists d such that $|xk - x| < d$ implies $|f(x') - f(x)| < \epsilon$. Since $x_n \to x$, there exists N such that when $n > N$ we have $|x_n - x| < d$. Thus, when $n > N$ we have $|f(x_n) - f(x)| < \epsilon$. Therefore, $f_n(x) \to f(x)$.

Conversely, assume for any $x_n \to x$ we have $f(x_n) \to f(x)$. Suppose f is not continuous at x. Then there exists a ϵ such that for any d there is a $x' \in (x - d, x + d)$ and $|f(x') - f(x)| \geq \epsilon$. Now we construct a convergent $x_n \to x$ as follows. Take $d_1 = 1$. By the assumption that f is not continuous at x, there exists $x_1 \in (x - d_1, x + d_1)$ such that $|f(x_1) - f(x)| \geq \epsilon$. Take $d_2 = \frac{1}{2}$ and there exists $|f(x_2) - f(x)| \geq \epsilon$. Continuing this process, we construct a convergent sequence $x_n \to x$ but $f(x_n)$ does not converges to $f(x)$. This is a contradiction. **QED**

The following theorem is an important characterization of continuous functions.

We denote by $f^{-1}(O)$ the preimage whose values fall under the set O. That is,

$$f^{-1}(O) = \{x \mid f(x) \in O\}$$

Theorem 1.5.2: A function f definded on E is continuous, if and only if, for any open set O, there exists an open set U such that

$$f^{-1}(O) = U \cap E$$

.

Proof: Suppose f is continuous. If O is empty, then take $U = \emptyset$ and clearly $f^{-1}(\emptyset) = \emptyset$. So we may assume O is non-empty. Let D be $f^{-1}(O)$. For any $x \in D$, since O is open there exists ϵ such that $(f(x) - \epsilon, f(x) + \epsilon) \subseteq O$. Since f is continuous, there exists δ_x such that whenever $x' \in (x - \delta_x, x + \delta_x)$ and $x' \in E$, we have $f(x') \in (f(x) - \epsilon, f(x) + \epsilon) \subseteq O$. So $(x - \delta_x, x + \delta) \cap E \subseteq D$.

Take $U = \bigcup_{x \in D}(x - \delta_x, x + \delta_x)$. Clealy, if $x \in D$ then $x \in U \cap E$. On the other hand, by the above arugments, if $x \in U \cap E$ then $x \in D$. Therefore, $U \cap E = D$.

Conversely, suppose for any open set O there exists an open set E such that $f^{-1}(O) = U \cap E$. We would like to show f is continuous. Consider any $x \in E$. By the assumption, for any ϵ there exists an open set U such that $f^{-1}((f(x) - \epsilon, f(x) + \epsilon)) = U \cap E$. Note that $x \in U$. Since U can be written as a countabe union of open intervals, there must exist an interval I that contains x and any $x' \in I$ has the property of $f(x') \in (f(x) - \epsilon, f(x) + \epsilon)$. Finally, since I is an open interval, there exists $d > 0$ such that $(x - d, x + d) \subseteq I$ and satisfies the continuity condition. **QED**

Theorem 1.5.3 (Extreme Value Theorem): A continuous function f on a bounded closed set has a maximum and a minimum. *Proof:* We show that f is bounded. So by Completeness Axiom, the image of f has a supremum m. If f cannot achieve c, then the continuous function $g(x) = \frac{1}{m-x}$ would be unbounded, which is a contradiction.

To prove f is bounded, let E be the domain of f and C be the image. By Theorem 1.5.2, since f is continuous, for a $\epsilon > 0$, any $f(x) \in C$ is associate with an open interval $I_x = (x - \delta_x, x + \delta_x)$ that satisfies the continuity condition. So the collection $\{I_x\}$ is an open cover of E. By Heine-Borel Theorem, there is a finite sub-cover $I_1, \ldots, I_n$. Take $M = \max\{|f(x_k)| + \epsilon\}_{k=1}^n$. For any $x \in E$, there is a I_k such that $x \in I_k$ and $|f(x)| - |f(x_k)| \leq |f(x) - f(x_k)| < \epsilon$. Therefore,

$$f(x) < |f(x)| < |f(x_k)| + \epsilon \leq M$$

. That is, f is bounded. **QED**

Theorem 1.5.4 (Intermediate Value Theorem): f is a continuous function on a bounded closed interval $[a, b]$. Assume $f(a) < f(b)$. Then for any $f(a) < c < f(b)$, there exists $x_0 \in (a, b)$ such that $f(x_0) = c$. *Proof:* We construct a sequence of closed intervals $[a_n, b_n]$ as follows. Take $a_1 = a$ and $b_1 = b$. Consider their middle point $m = \frac{a+b}{2}$. If $f(m) \leq c$, then take $a_2 = m$ and $b_2 = b_1$. Otherwise, take $a_2 = a_1$ and $b_2 = m$. Continuing this process, we obtain a sequence of closed intervals $[a_n, b_n]$ having the property that each $f(a_n) \leq c$ and each $f(b_n) > c$.

Now, by Nested Set Theorem, there exists $x_0 \in \bigcap_{n=1}^{\infty} [a_n, b_n]$. It is not difficult to show that $a_n \to x_0$ and $b_n \to x_0$. By Proposition 1.5.1, we have $f(a_n) \to f(x_0)$ and $f(b_n) \to f(x_0)$, since f is continuous.

Finally, since every $f(a_n) \leq c$ and $f(a_n) \to f(x_0)$, we have $f(x_0) \leq c$. On the other hand, since every $f(b_n) > c$ and $f(b_n) \to f(x_0)$, we have $f(x_0) \geq c$. Therefore, $f(x_0) = c$. **QED**

Uniform and Lipschitz Continuous

We say that f is a **uniformly continuous** function if, for any $\epsilon > 0$ there exists d such that for any x in the domain of x, whenever $|x' - x| < d$ we have $|f(x') - f(x)| < \epsilon$.

In other words, a function is uniformly continuous if it is continuous at every point "with the same pace".

A uniformly continuous function is clearly continuous. The reverse is not true. Consider the function $f(x) = \frac{1}{x}$ defined on $(0, 1)$. It is a continuous function. But for any x_1, x_2, if

$$|f(x_1) - f(x_2)| = \frac{|x_1 - x_2|}{x_1 x_2} < \epsilon$$

, then

$$|x_1 - x_2| < \epsilon x_1 x_2$$

. Since x_1, x_2 can be arbitrary small, there is no such uniform d.

A function is said to be **Lipschitz continuous** if there is a $c > 0$ such that for any x_1, x_2,

$$|f(x_1) - f(x_2)| < c \cdot |x_1 - x_2|$$

.

Clearly a Lipschitz continuous function is uniformly continuous. The reverse is , again, not true.

Theorem 1.5.5: A continuous function on a bounded closed set is uniformly continuous. *Proof:* Let f be a contious function that is defined on a bounded closed set E. Consider a ϵ. Let $B(x,d) = (x-d, x+d)$ be a neighbor of x with radius d. For any $x \in E$, since f is continuous the open set $B(f(x), \frac{\epsilon}{2})$ has an open pre-image $B(x,d)$. Note that the collection of open sets $\{B(x, \frac{d}{2})\}_{x \in E}$ is an open cover of E. Since E is bounded and closed, by Heine-Borel Theorem, E has a finite sub-cover $B(x_1, \frac{d_1}{2}), \ldots, B(x_k, \frac{d_k}{2})$.

Now take $h = \min\{\frac{d_1}{2}, \ldots, \frac{x_k}{2}\}$. Consider $x \in E$. Then $x \in B(x_i, \frac{d_i}{2})$ for some i. So $|f(x) - f(x_i)| < \frac{\epsilon}{2}$. When $|x' - x| < h$, we have $x' \in B(x_i, d_i)$. Thus, $|f(x') - f(x_i)| < \frac{\epsilon}{2}$. Therefore,

$$|f(x) - f(x')| < |f(x) - f(x_i)| + |f(x_i) - f(x)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

. The uniform h meets the challenge of ϵ, and f is uniformly continuous on E. **QED**

To see that a uniformly continuous function is not necessarily Lipschitz continuous, consider the function $f(x) = \sqrt{x}$ defined on $[0,1]$. By Theorem 1.5.5, f is uniformly continuous. If f is Lipschitz continuous, then there exists a c such that when $x = 0, y = \frac{1}{4c^2}$ we have

$$|y - x| = \frac{1}{2c} < c \cdot |\frac{1}{4c^2}| = \frac{1}{4c}$$

, which is a contradiction.